CP 218 Theory and Applications of Bayesian Learning

Project 1: Predict the connection between people

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1 Introduction

Online social networks can be represented as large graphs where nodes correspond to users and edges represent relationships between them. Link prediction is a fundamental task in network analysis that aims to estimate the likelihood of a connection between two nodes based on the observed structure of the network.

In this project, a partial crawl of the Twitter network is analysed where nodes represent users and directed edges represent follow relationships. The goal is to predict whether a link will exist between two users. To solve this problem, structural graph features are extracted from the network and used as inputs to supervised machine learning models. The models are evaluated using the ROC-AUC metric, which measures the ability to distinguish between existing and non-existing links.

2 Data Preparation

The training dataset is provided as an **adjacency list representing a directed Twitter network**. Each row contains a source user followed by the list of users they follow. This raw graph representation is converted into a dataset suitable for supervised learning by generating positive and negative edge samples.

2.1 Positive Edge Sampling

- A total of 40,000 edges were randomly sampled from the training network.
- These edges represent existing relationships between users and were labeled as **positive samples** ($label = 1$).

2.2 Negative Edge Sampling

- A total of 150,000 node pairs without an edge between them were randomly sampled.
- These pairs represent non-existing connections and were labeled as **negative samples** ($label = 0$).
- This sampling strategy creates a realistic imbalance similar to real social networks, where the number of non-existing connections greatly exceeds existing ones.

3 Feature Extraction

For each node pair (u, v), structural features are extracted from the graph to capture their connectivity patterns. These features describe how closely two users are related within the network structure. In total, **27 graph-based features** were generated, capturing node connectivity, neighborhood similarity, and classical link prediction measures.

3.1 Degree-based Features

These features describe the connectivity level of nodes.

Feature	Formula
Total Degree	$deg(u) = in_deg(u) + out_deg(u)$ $deg(v) = in_deg(v) + out_deg(v)$
In-degree Ratio	$\frac{in_deg(u)}{deg(u)}$
Out-degree Ratio	$\frac{out_deg(u)}{deg(u)}$

Nodes with higher degree values are more likely to form new connections, making degree-based features useful predictors for link formation.

3.2 Neighborhood Similarity Features

These features measure the similarity between the neighborhoods of nodes.

- Jaccard Similarity

$$J(u,v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}$$

Nodes that share many neighbors often belong to the same community and therefore have a higher probability of forming links.

3.3 Two-hop Connectivity Features

These features capture indirect relationships in the graph.

- Transitive paths $u \to x \to v$
- Reverse transitive paths $v \to x \to u$
- Total number of two-hop connections

Such paths indicate potential link formation through mutual connections.

3.4 Link Prediction Metrics

Several classical link prediction measures were also included.

Metric	Formula
Preferential Attachment	$Score = deg(u) \times deg(v)$
Adamic-Adar Index	$AA(u,v) = \sum_{z \in N(u) \cap N(v)} \frac{1}{\log(deg(z))}$
Resource Allocation Index	$RA(u,v) = \sum_{z \in N(u) \cap N(v)} \frac{1}{deg(z)}$

These metrics capture structural properties commonly observed in social networks.

3.5 Degree Difference Features

Feature	Formula
Total Degree Difference	$\lvert deg(u) - deg(v) \rvert$
In-degree Difference	$\lvert in_deg(u) - in_deg(v) \rvert$
Out-degree Difference	$\lvert out_deg(u) - out_deg(v) \rvert$

3.6 Additional Graph Features

Additional features include the in-degree of the destination node v and the out-degree of the source node u.

4 Model Training and Cross-Validation

4.1 Stratified K-Fold Cross-Validation

- Stratified 5-Fold Cross-Validation is used to evaluate model performance.
- The dataset is divided into five folds while preserving the ratio of positive and negative samples.
- In each iteration, four folds are used for training and one fold is used for validation.

4.2 Model Training

Three machine learning models were trained:

- Logistic Regression
- Random Forest
- XGBoost

Logistic Regression serves as a baseline model, while tree-based models such as Random Forest and XGBoost are capable of learning nonlinear relationships between graph features.

4.3 Evaluation Metric

Model performance is evaluated using **ROC-AUC**, which measures how well the model distinguishes between positive and negative classes.

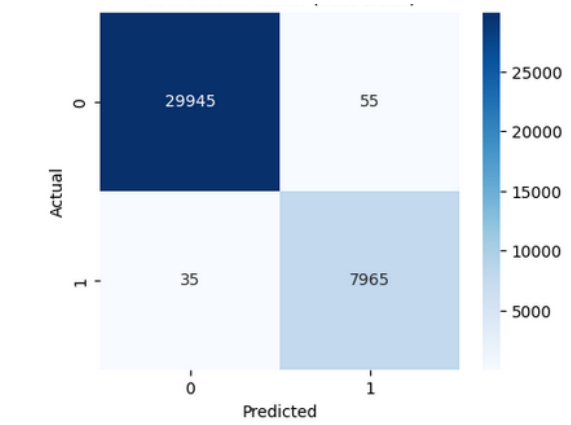


Figure 1: Confusion Matrix of Validation Dataset

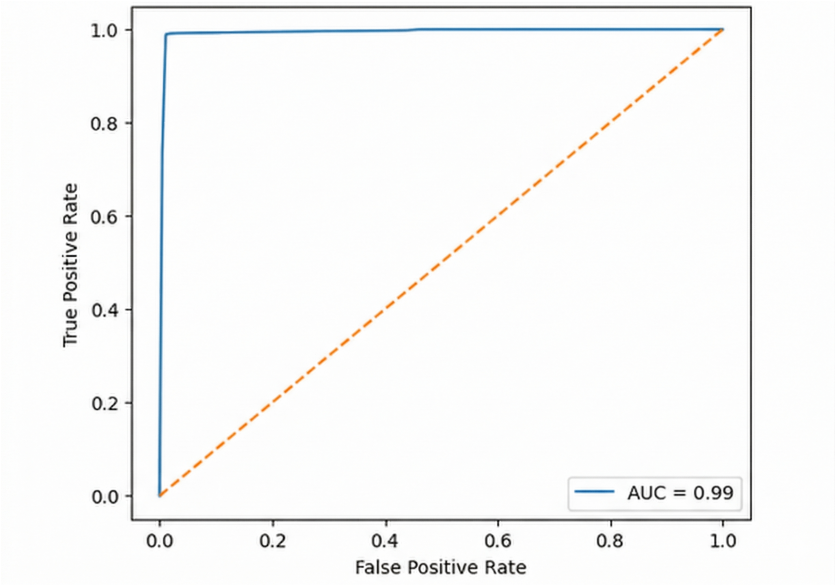


Figure 2: ROC Curve on Validation Dataset

4.4 Final Prediction and Submission

After selecting the best model based on cross-validation results, the model is retrained on the entire training dataset. The same feature extraction process is applied to the competition test dataset, and the model predicts the probability of link existence for each node pair. These predictions are saved in a `submission.csv` file for Kaggle evaluation.

5 Model Performance Analysis

Three models were evaluated using Stratified 5-Fold Cross-Validation: Logistic Regression, Random Forest, and XGBoost. Among these models, XGBoost achieved the highest average ROC-AUC score of approximately 0.99.

Logistic Regression assumes a linear relationship between features and the target variable. However, link prediction in networks often involves complex nonlinear structural relationships, which limits the performance of linear models.

Random Forest improves performance by modeling nonlinear relationships using multiple decision trees trained with bagging. Although this reduces variance and improves robustness, the trees are constructed independently and do not iteratively correct prediction errors.

XGBoost further improves performance by using gradient boosting, where trees are built sequentially and each new tree focuses on correcting the residual errors of previous trees. This allows the model to learn more complex decision boundaries between connected and non-connected node pairs.

XGBoost also incorporates regularization (ℓ_1 and ℓ_2) and optimized tree construction, which helps to control model complexity and improves generalization. These characteristics make it particularly suitable for high-dimensional tabular features derived from graph structures.

Feature importance analysis shows that graph structural features such as common neighbors, Jaccard similarity, Adamic-Adar index, and preferential attachment contributed strongly to the model's predictive performance. These features capture important connectivity patterns in social networks, where users sharing neighbors or exhibiting similar connectivity structures are more likely to form links.

6 Conclusion

This project addressed the link prediction problem by transforming graph relationships into a supervised learning task using structural graph features. A total of 27 features capturing connectivity patterns and neighborhood similarity were extracted from the network.Multiple models were evaluated using Stratified 5-Fold Cross-Validation with ROC-AUC as the evaluation metric. Among them, XGBoost achieved the best performance (AUC $\approx$ 0.99), demonstrating its effectiveness for link prediction in social networks.